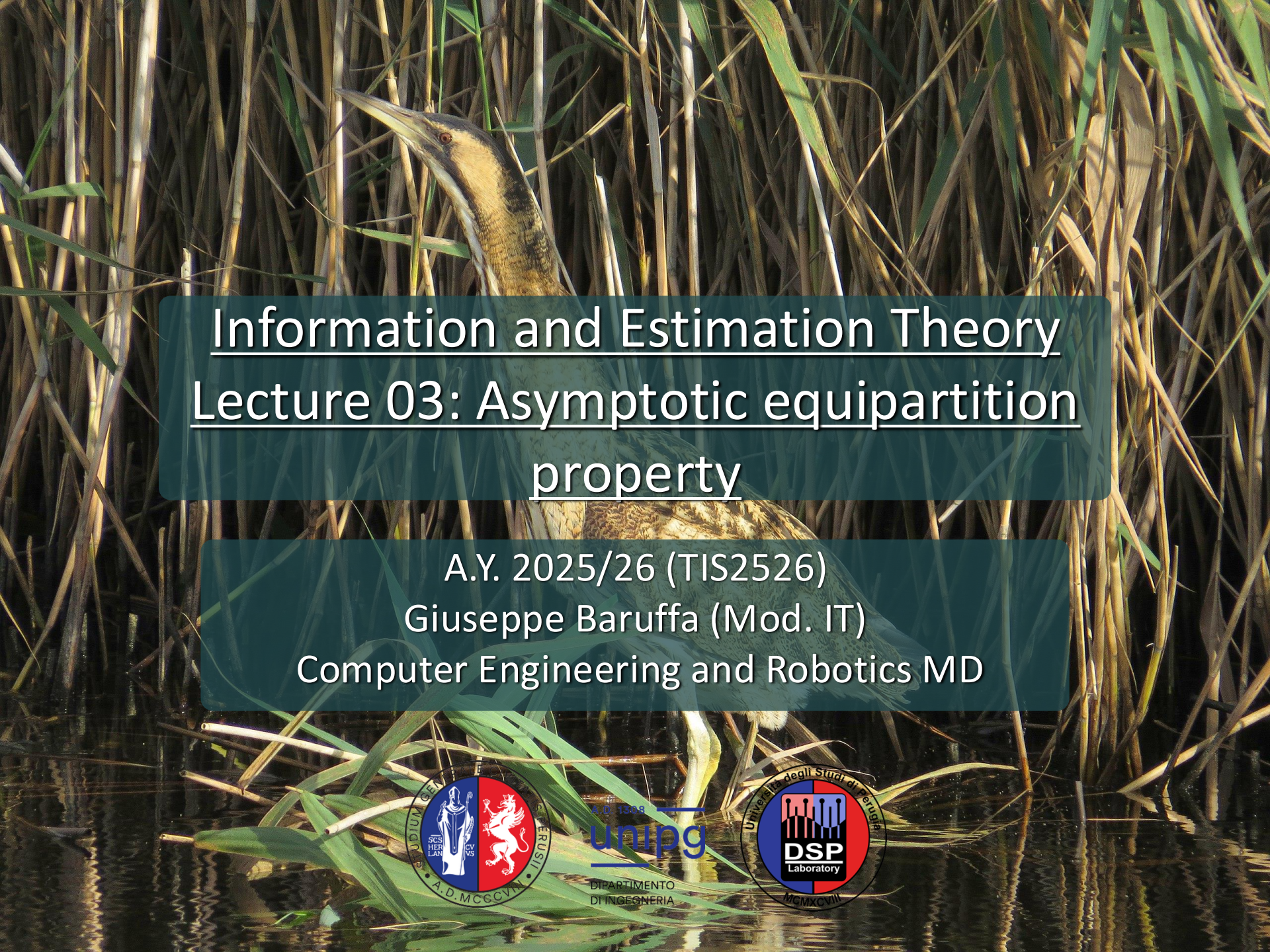


Information and Estimation Theory
Lecture 03: Asymptotic equipartition property

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Weak Law of Large Numbers (Bernoulli's Theorem)

- The **(weak) law of large numbers** , a.k.a. Bernoulli's theorem, states that, given n i.i.d. RVs X_i with mean μ and std σ, the new RV $X = \frac{1}{n}\sum_{i=1}^{n} X_i$ (*sample mean*) is close to the expected value $\mathrm{E}\{X_i\} = \mu$ for large values of n

- Indeed, $\mathrm{E}\{X\} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}\{X_i\} = \frac{n\mu}{n} = \mu$, and $\mathrm{E}\{(X-\mu)^2\} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$
- Using the Chebyshev's inequality, the probability that X differs from μ more than $\epsilon > 0$ is

$$\Pr\{|X-\mu| \geq \epsilon\} \leq \frac{\mathrm{E}\{(X-\mu)^2\}}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}$$

- Then, when $n \to \infty$,

$$\lim_{n\to\infty} \Pr\{|X-\mu| < \epsilon\} = 1 \quad \text{or} \quad \lim_{n\to\infty} \Pr\{|X-\mu| \geq \epsilon\} = 0$$

Asymptotic Equipartition Property

- The **asymptotic equipartition property (AEP)** is the IT analog of the law of large numbers
- The AEP states that

  $\frac{1}{n}\log\frac{1}{p(X^n)}$ is close to the entropy H, where $X^n = X_1, X_2, \ldots, X_n$ is a sequence of i.i.d. RVs and $p(X^n)$ is the probability of observing the sequence X^n

- Thus, $p(X^n)$ of an observed sequence will be close to 2^{-nH}
- This divides all the sequences into two sets, the **typical set**, where the sample entropy is close to the true entropy, and the **nontypical set**, which contains the other sequences
- Any property that is proved for the typical sequences will then be true (with high probability) and will determine the average behavior of a large sample

Example

- Let the RV $X \in \{0,1\}$ have a PMF defined by $p(1) = p$ and $p(0) = q$
- If X^n are i.i.d. $\sim p(x)$, the probability of a sequence $x^n = x_1, x_2, \ldots, x_n$ is $\Pr\{X^n = x^n\} = \prod_{i=1}^{n} p(x_i)$
  - E.g., the probability of sequence $x^6 = 1,0,1,1,0,1$ is $p^{\Sigma x_i} q^{6-\Sigma x_i} = p^4 q^2$
- Clearly, not all 2^n sequences of length n are equiprobable
- The probability of the observed sequence is the RV $p(X^n)$ of the outcomes X^n, i.i.d. $\sim p(x)$
  - Self-referential, but well-defined: we are asking for the probability of an event drawn according to the same probability distribution
- It turns out that $p(X^n)$ is close to 2^{-nH} with high probability

- "**Almost all events are almost equally surprising**", that is

$$\Pr\{(X_1, X_2, \ldots, X_n) : p(X_1, X_2, \ldots, X_n) = 2^{-n(H \pm \epsilon)}\} \approx 1$$

The probability of sequence X^n such that its probability is $2^{-n(H \pm \varepsilon)}$ is almost sure

Example

- Consider n=3, then we have the RV $p(X^3)$ which assumes the following values with probabilities (using p=0.7,q=0.3)

X^3	Value of $p(X^3)$	Pr{ $p(X^3)$ = value }
000	q^3=0.027	q^3=0.027
001	pq^2=0.063	$3pq^2$=0.189
010		
100		
011	p^2q=0.147	$3p^2q$=0.441
101		
110		
111	p^3=0.343	p^3=0.343

Convergence of RVs

- In the examples we are simply saying that the number of 1's in the sequence is close to *np* (with high probability), and all such sequences have (roughly) the same probability $2^{-nH(p)}$

- **Convergence of RVs:** A sequence of RVs X^n converges to a RV X
  1. $X_n \xrightarrow{\text{P}} X$ in probability if for every $\varepsilon > 0$, $\lim_{n\to\infty} \Pr\{|X_n - X| > \epsilon\} = 0$
  2. $X_n \xrightarrow{\text{L}^2} X$ in mean square sense if $\lim_{n\to\infty} \text{E}\{(X_n - X)^2\} = 0$
  3. $X_n \xrightarrow{\text{A.S.}} \text{X}$ with probability 1 (also almost surely) if $\Pr\left\{\lim_{n\to\infty} X_n = X\right\} = 1$

AEP Theorem

- **Theorem:** If X^n are i.i.d. $\sim p(x)$, then

$$-\frac{1}{n}\log p(X^n) \xrightarrow{\mathrm{P}} H(X)$$

  - *Proof:* Functions of independent RVs are also independent RVs. Thus, since the X_i are i.i.d., so are $\log p(X_i)$.
    Hence, by the weak law of large numbers,

$$-\frac{1}{n}\log p(X^n) = -\frac{1}{n}\log\prod_{i=1}^{n} p(X_i) = -\frac{1}{n}\sum_{i=1}^{n}\log p(X_i) \xrightarrow{\mathrm{P}} -\,\mathrm{E}\{\log p(X)\} = H(X)$$

Typical Set

- The **typical set $A_\varepsilon^{(n)}$** with respect to $p(x)$ is the set of sequences $x^n \in \mathcal{X}^n$ with the property

$$2^{-n(H(X)+\epsilon)} \leq p(x^n) \leq 2^{-n(H(X)-\epsilon)}$$

- **Theorem:** the typical set $A_\varepsilon^{(n)}$ has the following properties
  1. If $x^n \in A_\varepsilon^{(n)}$, then $H(X) - \varepsilon \leq -(1/n) \log p(x^n) \leq H(X) + \varepsilon$
  2. $\Pr\{A_\varepsilon^{(n)}\} > 1 - \varepsilon$ for n sufficiently large
  3. $|A_\varepsilon^{(n)}| \leq 2^{n(H(X) + \varepsilon)}$
  4. $|A_\varepsilon^{(n)}| \geq (1 - \varepsilon)2^{n(H(X) - \varepsilon)}$ for n sufficiently large

Thus, the typical set has probability nearly 1, all its elements are nearly equiprobable, and the number of its elements is nearly 2^{nH}

Proof of Typical Set Theorem

- *Proof:* (1) is immediate from the definition of $A_\varepsilon^{(n)}$
- (2) follows from the AEP Theorem, since $\Pr\{X^n \in A_\varepsilon^{(n)}\}$ tends to 1 as $n \to \infty$. Thus, for any $\delta > 0$, there exists an n_0 such that for all $n \geq n_0$, we have

$$\Pr\left\{\left|-\frac{1}{n}\log p(X^n) - H(X)\right| < \epsilon\right\} > 1 - \delta$$

  Setting $\delta = \varepsilon$, we obtain the second part of the theorem

- To prove (3)

$$1 = \sum_{x^n \in \mathcal{X}^n} p(x^n) \geq \sum_{x^n \in A_\epsilon^{(n)}} p(x^n) \geq \sum_{x^n \in A_\epsilon^{(n)}} 2^{-n(H(X)+\epsilon)} = 2^{-n(H(X)+\epsilon)} \left|A_\epsilon^{(n)}\right|$$

from definition of typical set

$$\left|A_\epsilon^{(n)}\right| \leq 2^{n(H(X)+\epsilon)}$$

- For (4), for sufficiently large n, $\Pr\{A_\varepsilon^{(n)}\} > 1 - \varepsilon$, so that

$$1 - \epsilon < \Pr\left\{A_\epsilon^{(n)}\right\} \leq \sum_{x^n \in A_\epsilon^{(n)}} 2^{-n(H(X)-\epsilon)} = 2^{-n(H(X)-\epsilon)} \left|A_\epsilon^{(n)}\right|$$

from definition of typical set

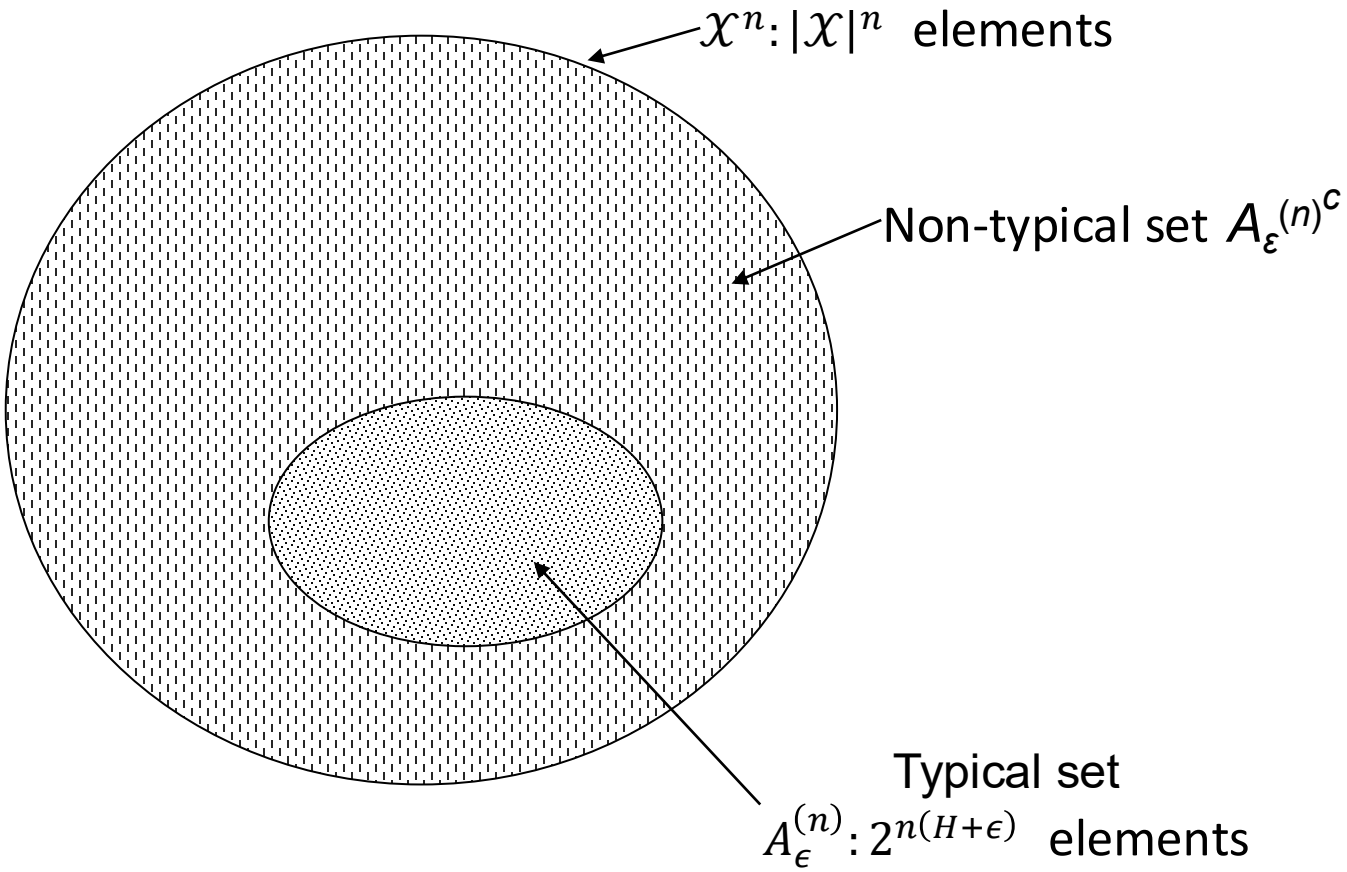
$$\left|A_\epsilon^{(n)}\right| \geq (1-\epsilon)2^{n(H(X)-\epsilon)}$$

Example of Typical Set

- Let $n = 4$, $X \sim \text{Bern}(p)$, $p = 0.4$, $\mathcal{X} = \{0,1\}$, $\epsilon = 0.1$, then $H(X) = 0.971$
- A sequence is typical if $\left|-\frac{1}{4}\log p(x^4) - H(X)\right| < 0.1$
- All sequences with roughly $np = 4 \cdot 0.4 = 1.6$ ones should be in $A_{0.1}^{(4)}$
  - 0 ones: $p(0,0,0,0) = \Pr(X^4 = 0,0,0,0) = 0.4^0 0.6^4 = 0.1296$
    - Most probable, but not typical
    - $\left|-\frac{1}{4}\log p(x^4) - H(X)\right| = |0.737 - 0.971| = 0.234 > 0.1$ → NO
  - 1 one: $p(x^4) = \Pr(X^4 = x^4 : \sum_i x_i = 1) = 0.4^1 0.6^3 = 0.0864$
    - $|0.883 - 0.971| = 0.088 < 0.1$ → YES, there are $\binom{4}{1} = 4$ such sequences
  - 2 ones: $p(x^4) = 0.4^2 0.6^2 = 0.0576$
    - $|1.030 - 0.971| = 0.059 < 0.1$ → YES, there are $\binom{4}{2} = 6$ such sequences
  - 3-4 ones: $p(x^4) = 0.4^3 0.6^1 = 0.0384$ and $p(x^4) = 0.4^4 0.6^0 = 0.0256$
    - $|1.176 - 0.971| = 0.205 > 0.1$ and $|1.321 - 0.971| = 0.350 > 0.1$ → NO
- $A_{0.1}^{(4)} = \{0001,0010,0100,1000,0011,0101,0110,1001,1010,1100\}$
- $\left|A_{0.1}^{(4)}\right| = 10 \leq 2^{4(0.971+0.1)} = 19.52$

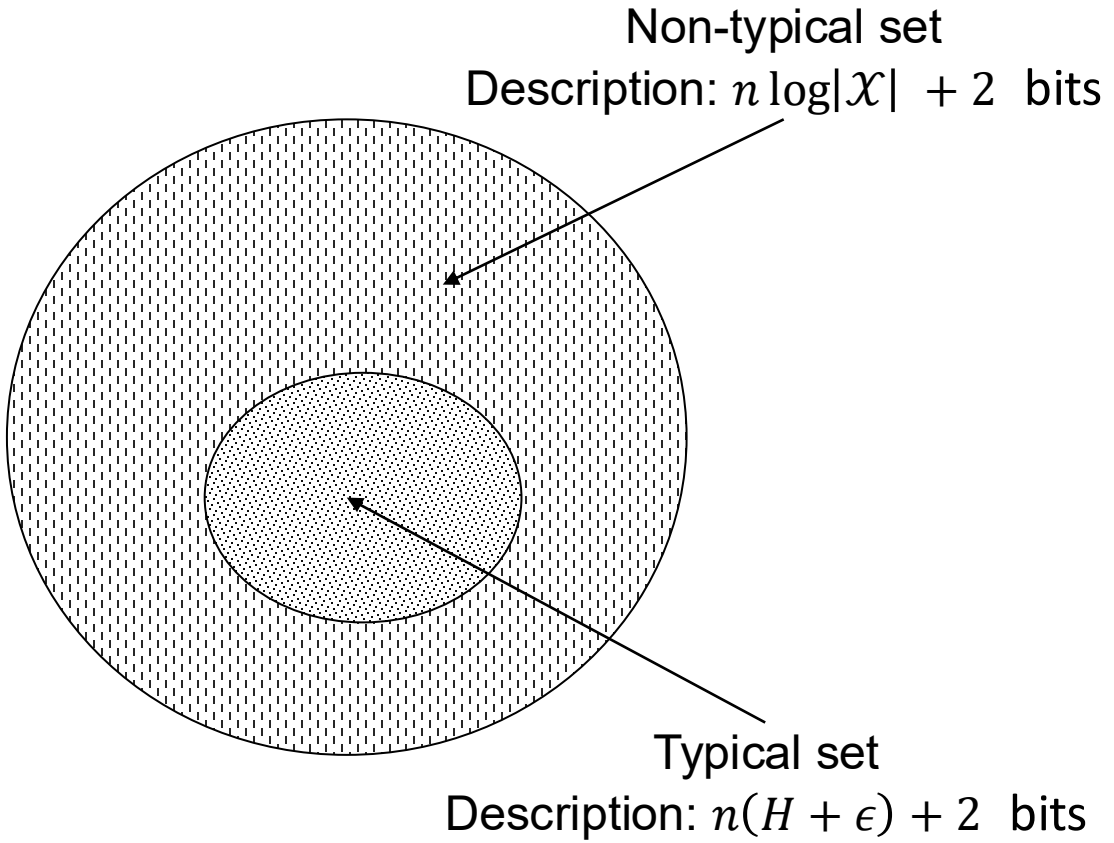
Consequence of AEP: Data Compression

- We wish to find short descriptions for sequences X^n of RVs i.i.d. $\sim p(x)$
- Divide into 2 sets: the typical set $A_\varepsilon^{(n)}$ and its complement $A_\varepsilon^{(n)^C}$



Consequence of AEP: Data Compression

- Order all elements in each set in some way (e.g., lexicographic)
- Represent each sequence of $A_\varepsilon^{(n)}$ with its index in the set
- Since there are $\leq 2^{n(H+\varepsilon)}$ sequences in $A_\varepsilon^{(n)}$, the indexing requires no more than $n(H+\varepsilon)+1$ bits
  - The extra bit may be necessary because $n(H+\varepsilon)$ may not be an integer
- We prefix all these sequences by a "0", giving a total length of $\leq n(H+\varepsilon)+2$ bits to represent each sequence in $A_\varepsilon^{(n)}$



Consequence of AEP: Data Compression

- We index each seq. not in $A_\varepsilon^{(n)}$ by using $\leq n \log|\mathcal{X}|+1$ bits
- Prefixing these indices by "1", we have a code for seqs. in $\mathcal{X}^n$

- Note the following features of the above coding scheme
  - The code is **one-to-one** and easily decodable
    - The initial bit acts as a flag bit to indicate the length of the codeword that follows
  - Brute-force enumeration of the **atypical set $A_\varepsilon^{(n)c}$**, without considering $|A_\varepsilon^{(n)c}| < |\mathcal{X}^n|$, surprisingly, yields an efficient description
  - The typical sequences have short descriptions of length $\approx nH$

Consequence of AEP: Data Compression

- $l(x^n)$ is the codeword length of sequence x^n
- If n is sufficiently large so that $\Pr\{A_\varepsilon^{(n)}\} \geq 1 - \varepsilon$, expected $l(x^n)$ is

$$E\{l(X^n)\} = \sum_{x^n \in \mathcal{X}^n} p(x^n) l(x^n) = \sum_{x^n \in A_\epsilon^{(n)}} p(x^n) l(x^n) + \sum_{x^n \in A_\epsilon^{(n)c}} p(x^n) l(x^n)$$

$$\leq \sum_{x^n \in A_\epsilon^{(n)}} p(x^n)(n(H+\epsilon)+2) + \sum_{x^n \in A_\epsilon^{(n)c}} p(x^n)(n \log|\mathcal{X}| + 2)$$

$$= \overset{P}{\Pr\left\{A_\epsilon^{(n)}\right\}}(n(H+\epsilon)+2) + \overset{1-P}{\Pr\left\{A_\epsilon^{(n)c}\right\}}(n \log|\mathcal{X}| + 2)$$

$Pn(H+\varepsilon) + 2P + (1-P)\, n \log|\mathcal{X}| + 2(1-P) = Pn(H+\varepsilon) + (1-P)\, n \log|\mathcal{X}| + 2$, $P<1$ and $(1-P)<\varepsilon$

$$\leq n(H+\epsilon) + \epsilon n \log|\mathcal{X}| + 2 = n(H+\epsilon')$$

- $\varepsilon' = \varepsilon + \varepsilon \log|\mathcal{X}| + 2/n$ can be made arbitrarily small by an appropriate choice of ε followed by an appropriate choice of n

Data Compression Theorem

- **Theorem:** Let X^n be i.i.d. $\sim p(x)$ and $\varepsilon > 0$. Then there exists a code that maps sequences x^n of length n into binary strings such that the mapping is **one-to-one** (and therefore invertible) and, for n sufficiently large,

$$\mathrm{E}\left\{\frac{1}{n} l(X^n)\right\} \leq H(X) + \epsilon$$

  - *Proof:* just proved in the previous slides

- Thus, we can represent sequences X^n using $nH(X)$ bits (on the average)

High Probability Sets

- From the definition of $A_\varepsilon^{(n)}$, it is a small set that contains most of the probability
  - But it is not clear whether it is the smallest such set
- $\forall\ n = 1,2,\ldots$, let $B_\delta^{(n)} \subset \mathcal{X}^n$ be the smallest set with $\Pr\{B_\delta^{(n)}\} \geq 1 - \delta$
  - $B_\delta^{(n)}$ must have significant intersection with $A_\varepsilon^{(n)}$ and therefore must have about as many elements

- **Theorem (w/o proof):** Let $X_1, X_2, \ldots, X_n$ be i.i.d.$\sim p(x)$. For $\delta < 1/2$ and any $\delta' > 0$, if $\Pr\{B_\delta^{(n)}\} > 1 - \delta$, then

$$\frac{1}{n}\log\left|B_\delta^{(n)}\right| > H - \delta' \qquad \text{for } n \text{ sufficiently large}$$

  Thus, $B_\delta^{(n)}$ must have at least 2^{nH} elements, <u>to first order in the exponent</u>
  - But $A_\varepsilon^{(n)}$ has $2^{n(H \pm \varepsilon)}$ elements, therefore it is about the same size as the smallest high-probability set

Example

- With the previous example for typical sequences, let $\delta = 0.1$
- Find the smallest set such that $\Pr\left\{B_{0.1}^{(4)}\right\} \geq 0.9$
  - 0 ones: $p(0,0,0,0) = \Pr(X^4 = 0,0,0,0) = 0.4^0 0.6^4 = 0.1296$
  - 1 ones: 4 sequences, we reach 0.1296+4·0.0864=0.4752
  - 2 ones: 6 sequences, we get 0.4752+6·0.0576=0.8208
  - 3 ones: add 3 out of 4 sequences, to achieve 0.8208+3·0.0384=0.936
  - $B_{0.1}^{(4)}$ ={0000,
    0001,0010,0100,1000,
    0011,0101,0110,1001,1010,1100,
    0111,1011,1101}
  - So it has cardinality 14, whereas the typical set has 10

First Order in the Exponent

- The notation $a_n \doteq b_n$ means $\lim_{n\to\infty} \frac{1}{n} \log \frac{a_n}{b_n} = 0$,
  i.e., equal to the 1st order in the exponent
- So, we restate that if $\delta_n \to 0$ and $\varepsilon_n \to 0$, then

$$|B_{\delta_n}{}^{(n)}| \doteq |A_{\varepsilon_n}{}^{(n)}| \doteq 2^{nH}$$

$$(1-\epsilon_n)2^{nH}2^{-n\epsilon_n} \le \left|A_{\epsilon_n}^{(n)}\right| \le 2^{nH}2^{n\epsilon_n}$$

$$\log(1-\epsilon_n) + \log 2^{nH} - \log 2^{n\epsilon_n} \le \log\left|A_{\epsilon_n}^{(n)}\right| \le \log 2^{nH} + \log 2^{n\epsilon_n}$$

$$\log(1-\epsilon_n) - \log 2^{n\epsilon_n} \le \log\left|A_{\epsilon_n}^{(n)}\right| - \log 2^{nH} \le n\epsilon_n$$

$$\frac{1}{n}\log(1-\epsilon_n) - \epsilon_n \le \frac{1}{n}\log\frac{\left|A_{\epsilon_n}^{(n)}\right|}{2^{nH}} \le \epsilon_n$$

$$\lim_{n\to\infty}\frac{1}{n}\log\frac{\left|A_{\epsilon_n}^{(n)}\right|}{2^{nH}} = 0$$